PULSEIQ — PHASE 1 REPORT

Data Analytics | Dilip D | 24 October 2026

Repository: https://github.com/Dilip111004/pulseiq **Git tag:** phase-1

1. Project summary

PulseIQ is a workforce analytics and data-governance project covering CSV ETL and validation, warehouse reconciliation, SQL data-quality checks, governed metric views, and a Streamlit dashboard. This report records the available implementation evidence and clearly identifies remaining sign-off items.

2. Five-task status and evidence

Task	Status	Evidence link
PQ-01 — ETL & contracts	Verified	Task-by-task guide; 29 CSVs / 62,770 rows; 62,732 accepted, 38 quarantined.
PQ-02 — Warehouse	Verified	Row reconciliation; 25 tables reconciled; 39 FK constraints recorded.
PQ-03 — Data quality	Verified	Mutation evidence; all 14 SQL-backed mutation checks detected introduced issues in a separate test DB.
PQ-04 — Metrics	Verified with limitation	Metric hand-checks; 28 views executed, six independent checks matched; M28 not measurable yet.
PQ-05 — Dashboard	Implemented; review pending	Streamlit app; four KPI cards, five charts, filters; screenshots.

3. Proof it works

• Automated tests: 13 passed in the recorded run (python -m pytest -q).
• ETL repeatability: 58 clean/rejected outputs were byte-identical across two runs; row accounting passed.
• Warehouse: 25 tables reconciled; zero orphan rows in five representative joins.
• Data quality: 14 SQL-backed mutation tests detected deliberately introduced defects in a separate test database; baseline preserved.
• Metric layer: 28 views executed and six independent hand-checks matched.

Additional evidence: DQ results · SQL DQ results · Metric checks

4. What I learned

I learned how to build a repeatable ETL workflow and preserve rejected records with reasons instead of silently dropping them. I learned to validate warehouse relationships and reconcile results independently. Mutation testing helped me verify that data-quality rules detect known defects. I also learned that dashboards need clearly defined metrics and trustworthy sources. When numbers do not reconcile, I should document evidence and ask the owner rather than guess at a correction.

5. Problems encountered and how they were handled

Payroll-to-ledger difference: June 2026 payroll employer cost was INR 54,664,233.35 versus ledger salary-related expense of INR 54,682,683.35: an unresolved INR 18,450 difference. It is traced to department D01, ledger line JL-005614, journal JE-002094, account 5001. I preserved the data and documented the discrepancy; Finance confirmation is still required before any correction or approved exception.

Freshness and metric limits: Five attrition predictions and 405 anomaly scores were stale relative to the recorded run. M28 cannot yet be measured because actuals and forecast periods do not overlap. These are recorded as limitations, not successful outcomes.

Reason-code contract: Eleven distinct reason-code values were observed, but no official approved allow-list was provided. Owner confirmation is required before treating the observed values as authoritative.

Dashboard review: The dashboard was visually checked and filters were tested, but metric calculations and source compliance still need final validation.

6. Plan for the next phase

1. Obtain Finance's explanation and approved resolution for the INR 18,450 difference; rerun reconciliation and retain evidence.
2. Obtain the official anomaly reason-code allow-list and validate against it.
3. Audit every dashboard metric against governed definitions, source tables, privacy thresholds, and filters; rerun tests after changes.
4. Address data freshness and measure M28 when actuals overlap the forecast period.
5. Refresh documentation, run a fresh test suite, verify the phase-1 tag points to the intended commit, and package next-phase deliverables.

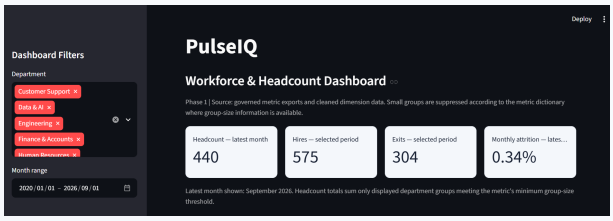
7. Sign-off status

Implementation evidence is available, but final sign-off remains conditional on the payroll reconciliation, official reason-code confirmation, and dashboard metric/source-compliance review. The 13-test result is from the recorded run and should be refreshed after code changes.

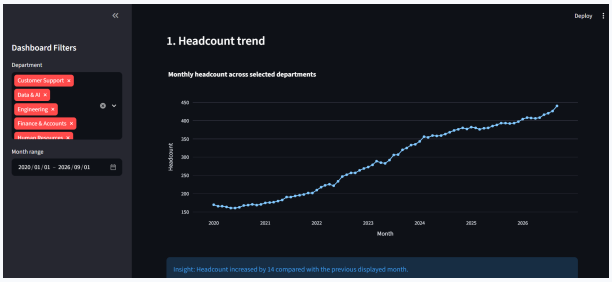
Links in this report point to the repository's master branch evidence files. The phase-1 tag link is included as reported by the author.

Dashboard Screenshots

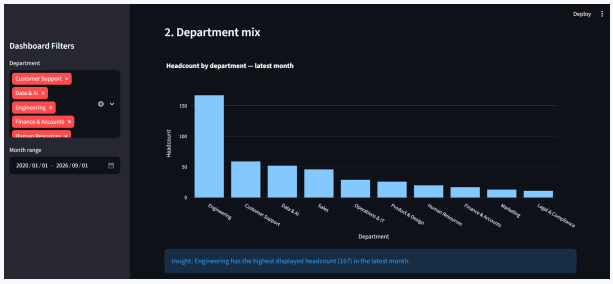
Screenshots of the dashboard and its five charts



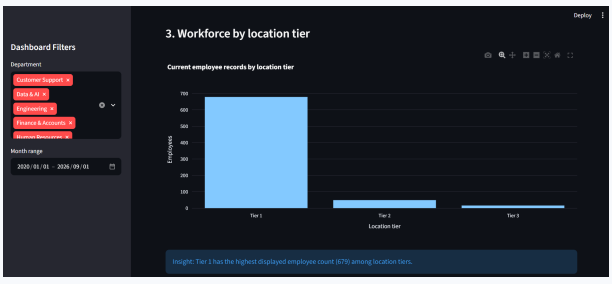
Dashboard overview



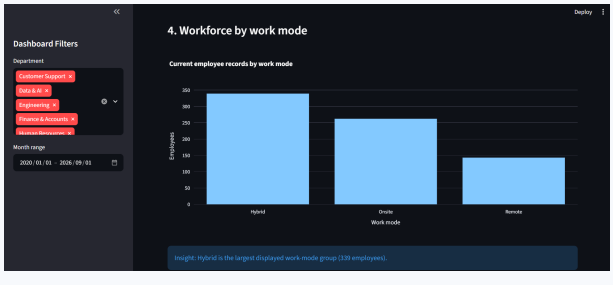
Headcount trend



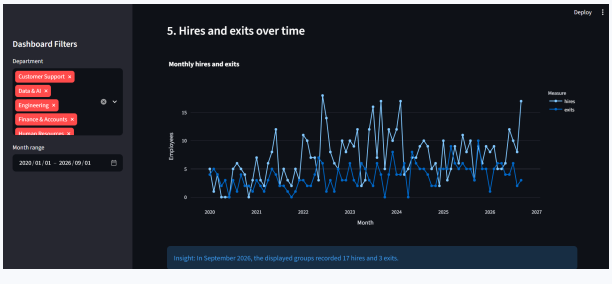
Department mix



Workforce by location tier



Workforce by work mode



Hires and exits